



Chapter 8: Area and Volume

Three-week revision booklet • 6 worksheets • 240 questions

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) • MA4-ARE-C-01 (area) • MA4-VOL-C-01 (volume and capacity) • MAO-WM-01

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How to use this booklet

There are six worksheets, two for each week. Each one drills a single skill from the Area and Volume topic test. Each sheet has the same shape: **40 questions**, split into ● **Easy**, ● **Medium** and ● **Challenging**. Work from the top of the page down — the questions get harder as you go. Show your working for the challenging questions. The final worksheet mixes every skill together, so it is the best check that you are ready for the topic test.

Week	Worksheet	Skill
1	1	Perimeter
1	2	Circumference
2	3	Area of shapes
2	4	Units and conversions
3	5	Volume and capacity
3	6	Mixed revision — all skills

Progress tracker

Worksheet	Date completed	Score / 40	Checked by
1 Perimeter			
2 Circumference			
3 Area of shapes			
4 Units and conversions			
5 Volume and capacity			
6 Mixed revision — all skills			

Worksheet 1: Perimeter

Week 1 · Worksheet 1 of 6 · 40 questions

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) · MA4-ARE-C-01 (area) · MA4-VOL-C-01 (volume and capacity) · MAO-WM-01

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Perimeter is the distance all the way around a shape. Add every side length.

EASY

20 QUESTIONS

Find the perimeter of each rectangle.

1. 15 cm by 9 cm _____
2. 8 m by 5 m _____
3. 12 mm by 7 mm _____
4. 20 cm by 4 cm _____
5. 6.5 m by 3 m _____
6. 14 cm by 14 cm _____
7. 25 mm by 10 mm _____
8. 9 m by 2.5 m _____
9. 18 cm by 11 cm _____
10. 7 m by 7 m _____

Find the perimeter of each shape.

11. Square, side 6 cm _____
12. Square, side 12 m _____
13. Equilateral triangle, side 9 cm _____
14. Triangle 5, 7, 8 cm _____
15. Regular pentagon, side 4 cm _____
16. Regular hexagon, side 3 m _____
17. Triangle 12, 16, 20 mm _____
18. Square, side 2.5 m _____
19. Rhombus, side 11 cm _____
20. Regular octagon, side 5 mm _____

MEDIUM

12 QUESTIONS

Answer each question.

21. A rectangle has perimeter 34 cm and length 11 cm. Find its width. _____
22. A square has perimeter 48 m. Find its side. _____
23. An equilateral triangle has perimeter 27 cm. Find its side. _____
24. A rectangle is twice as long as it is wide, perimeter 36 cm. Find both sides. _____
25. A rectangle has perimeter 50 m and width 8 m. Find its length. _____
26. An L-shape has sides 10, 4, 6, 5, 4, 9 cm. Find its perimeter. _____
27. A square has perimeter 30 cm. Find its side. _____
28. A triangle has perimeter 24 cm; two sides are 9 cm and 7 cm. Find the third. _____
29. A rectangle is 12 m by 7 m. Find the perimeter. _____
30. A regular hexagon has perimeter 42 cm. Find its side. _____
31. A rectangle has perimeter 40 cm and length 12 cm. Find its width. _____
32. A paddock is 45 m by 28 m. Find the fencing needed. _____

Worksheet 1: Perimeter

Week 1 • Worksheet 1 of 6 • 40 questions • continued

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) • MA4-ARE-C-01 (area) • MA4-VOL-C-01 (volume and capacity) • MAO-WM-01

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● CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

1. A rectangular paddock 45 m by 28 m needs fencing with a 3 m gate. Find the fencing needed.

.....

2. A garden bed is a square of side 4.5 m. Edging costs \$12 per metre. Find the cost.

.....

3. An L-shaped room has outer sides 8 m and 6 m, with a 3 m by 2 m corner removed. Find its perimeter.

.....

4. A rectangle has perimeter 60 cm. Its length is 6 cm more than its width. Find both.

.....

5. A square and a rectangle both have perimeter 24 cm. The rectangle is 8 cm long. Compare their widths.

.....

6. Fencing costs \$18/m. A 15 m by 9 m yard is fenced. Find the total cost.

.....

7. A triangle has perimeter 30 cm and is isosceles with equal sides 11 cm. Find the base.

.....

8. Explain why doubling every side of a shape doubles its perimeter.

.....

Worksheet 2: Circumference

Week 1 • Worksheet 2 of 6 • 40 questions

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) • MA4-ARE-C-01 (area) • MA4-VOL-C-01 (volume and capacity) • MAO-WM-01

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$C = 2\pi r$ or $C = \pi d$. Give answers correct to one decimal place unless told otherwise.

EASY

20 QUESTIONS

Find the circumference. The radius is given.

- | | | | |
|-----------------|-----------------|----------------|---------------|
| 1. 5 cm _____ | 2. 3 m _____ | 3. 10 mm _____ | 4. 7 cm _____ |
| 5. 2.5 m _____ | 6. 12 mm _____ | 7. 6 cm _____ | 8. 1 m _____ |
| 9. 8.5 cm _____ | 10. 20 mm _____ | | |

Find the circumference. The diameter is given.

- | | | | |
|------------------|------------------|---------------|-----------------|
| 11. 18 mm _____ | 12. 10 cm _____ | 13. 6 m _____ | 14. 24 mm _____ |
| 15. 4.2 cm _____ | 16. 50 mm _____ | 17. 9 m _____ | 18. 15 cm _____ |
| 19. 3.6 m _____ | 20. 100 mm _____ | | |

MEDIUM

12 QUESTIONS

Answer each question, correct to one decimal place.

- | | | |
|---|---|---|
| 21. Radius 6 cm _____ | 22. Diameter 14 cm _____ | 23. Radius 0.5 m _____ |
| 24. Diameter 7 mm _____ | 25. Radius 11 cm _____ | 26. Diameter 2.4 m _____ |
| 27. Radius 25 mm _____ | 28. Diameter 30 cm _____ | 29. A circle has radius 9 m. Find C.
_____ |
| 30. A circle has diameter 8.4 cm. Find C. _____ | 31. A wheel has radius 35 cm. Find C. _____ | 32. A plate has diameter 22 cm. Find C. _____ |

CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

- | | |
|--|--|
| 33. A circle has circumference 31.4 cm. Find its diameter.
..... | 34. A circle has circumference 62.8 m. Find its radius.
..... |
| 35. A bicycle wheel has diameter 66 cm. How far does it travel in one turn?
..... | 36. A wheel of radius 35 cm turns 10 times. How far does it travel?
..... |
| 37. A circular pond has radius 4 m. Find the length of edging needed.
..... | 38. Find the perimeter of a semicircle with diameter 12 cm.
..... |
| 39. Find the perimeter of a quarter circle with radius 8 cm.
..... | 40. Explain why $C = \pi d$ and $C = 2\pi r$ give the same answer.
..... |

Worksheet 3: Area of shapes

Week 2 • Worksheet 3 of 6 • 40 questions

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) • MA4-ARE-C-01 (area) • MA4-VOL-C-01 (volume and capacity) • MAO-WM-01

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Rectangle $A = lw$ • triangle $A = \frac{1}{2}bh$ • parallelogram $A = bh$ • circle $A = \pi r^2$.

EASY

20 QUESTIONS

Find the area of each rectangle.

- | | | | |
|------------------------|--------------------------|-------------------------|------------------------|
| 1. 8 cm by 5 cm _____ | 2. 12 m by 3 m _____ | 3. 10 mm by 10 mm _____ | 4. 15 cm by 4 cm _____ |
| 5. 7 m by 6 m _____ | 6. 20 cm by 2.5 cm _____ | 7. 9 mm by 11 mm _____ | 8. 6.4 m by 5 m _____ |
| 9. 14 cm by 3 cm _____ | 10. 25 m by 8 m _____ | | |

Find the area of each triangle (base, height).

- | | | | |
|------------------------|-----------------------|-----------------------|-----------------------|
| 11. 12 mm, 9 mm _____ | 12. 10 cm, 6 cm _____ | 13. 8 m, 5 m _____ | 14. 14 cm, 4 cm _____ |
| 15. 20 mm, 7 mm _____ | 16. 9 m, 6 m _____ | 17. 16 cm, 5 cm _____ | 18. 7 m, 4 m _____ |
| 19. 24 mm, 10 mm _____ | 20. 11 cm, 8 cm _____ | | |

MEDIUM

12 QUESTIONS

Find the area.

- | | | |
|---|--|--|
| 21. Parallelogram, base 20 m, height 7 m _____ | 22. Parallelogram, base 12 cm, height 9 cm _____ | 23. Circle, radius 6 cm (1 dp) _____ |
| 24. Circle, radius 3 m (1 dp) _____ | 25. Circle, diameter 10 cm (1 dp) _____ | 26. Circle, diameter 14 mm (1 dp) _____ |
| 27. Square, side 13 cm _____ | 28. Rectangle, 4.5 m by 6 m _____ | 29. Triangle, base 15 mm, height 12 mm _____ |
| 30. Parallelogram, base 8 m, height 4.5 m _____ | 31. Circle, radius 2.5 m (1 dp) _____ | 32. Triangle, base 6.4 cm, height 5 cm _____ |

CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

- | | |
|--|---|
| 33. A rectangular lawn 12 m by 8 m has a 3 m square bed removed. Find the lawn area left.
_____ | 34. An L-shape is a 10 cm by 6 cm rectangle with a 4 cm by 3 cm corner removed. Find its area.
_____ |
| 35. A triangle has area 48 cm^2 and base 12 cm. Find its height.
_____ | 36. A rectangle has area 84 m^2 and width 7 m. Find its length.
_____ |
| 37. A circle has area 78.5 cm^2 . Estimate its radius.
_____ | 38. A shape is a 12 cm by 5 cm rectangle with a semicircle of diameter 5 cm on one end. Find its area.
_____ |
| 39. Carpet costs $\$45/\text{m}^2$. A room is 4.5 m by 3.2 m. Find the cost.
_____ | 40. Explain why the area of a triangle is half that of a rectangle with the same base and height.
_____ |

Worksheet 4: Units and conversions

Week 2 · Worksheet 4 of 6 · 40 questions

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) · MA4-ARE-C-01 (area) · MA4-VOL-C-01 (volume and capacity) · MAO-WM-01

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$1 \text{ cm}^2 = 100 \text{ mm}^2$ · $1 \text{ m}^2 = 10\,000 \text{ cm}^2$ · $1 \text{ cm}^3 = 1000 \text{ mm}^3$ · $1 \text{ L} = 1000 \text{ mL}$ · $1 \text{ cm}^3 = 1 \text{ mL}$.

EASY

20 QUESTIONS

Complete each area conversion.

- | | | | |
|--|--|--|--|
| 1. $3 \text{ m}^2 = \underline{\hspace{1cm}} \text{ cm}^2$ | 2. $200 \text{ mm}^2 = \underline{\hspace{1cm}} \text{ cm}^2$ | 3. $5 \text{ cm}^2 = \underline{\hspace{1cm}} \text{ mm}^2$ | 4. $40\,000 \text{ cm}^2 = \underline{\hspace{1cm}} \text{ m}^2$ |
| 5. $7 \text{ m}^2 = \underline{\hspace{1cm}} \text{ cm}^2$ | 6. $900 \text{ mm}^2 = \underline{\hspace{1cm}} \text{ cm}^2$ | 7. $12 \text{ cm}^2 = \underline{\hspace{1cm}} \text{ mm}^2$ | 8. $80\,000 \text{ cm}^2 = \underline{\hspace{1cm}} \text{ m}^2$ |
| 9. $2.5 \text{ m}^2 = \underline{\hspace{1cm}} \text{ cm}^2$ | 10. $150 \text{ mm}^2 = \underline{\hspace{1cm}} \text{ cm}^2$ | | |

Complete each volume or capacity conversion.

- | | | | |
|---|---|---|--|
| 11. $15 \text{ cm}^3 = \underline{\hspace{1cm}} \text{ mm}^3$ | 12. $6000 \text{ mm}^3 = \underline{\hspace{1cm}} \text{ cm}^3$ | 13. $3 \text{ L} = \underline{\hspace{1cm}} \text{ mL}$ | 14. $2500 \text{ mL} = \underline{\hspace{1cm}} \text{ L}$ |
| 15. $8 \text{ cm}^3 = \underline{\hspace{1cm}} \text{ mL}$ | 16. $4 \text{ L} = \underline{\hspace{1cm}} \text{ cm}^3$ | 17. $9000 \text{ mm}^3 = \underline{\hspace{1cm}} \text{ cm}^3$ | 18. $0.5 \text{ L} = \underline{\hspace{1cm}} \text{ mL}$ |
| 19. $20 \text{ cm}^3 = \underline{\hspace{1cm}} \text{ mm}^3$ | 20. $750 \text{ mL} = \underline{\hspace{1cm}} \text{ L}$ | | |

MEDIUM

12 QUESTIONS

Complete each conversion.

- | | | |
|---|---|--|
| 21. $2 \text{ m}^2 = \underline{\hspace{1cm}} \text{ mm}^2$ | 22. $1\,000\,000 \text{ cm}^3 = \underline{\hspace{1cm}} \text{ m}^3$ | 23. $6.5 \text{ L} = \underline{\hspace{1cm}} \text{ mL}$ |
| 24. $35 \text{ cm}^3 = \underline{\hspace{1cm}} \text{ L}$ | 25. $0.4 \text{ m}^2 = \underline{\hspace{1cm}} \text{ cm}^2$ | 26. $250 \text{ cm}^3 = \underline{\hspace{1cm}} \text{ mL}$ |
| 27. $12 \text{ L} = \underline{\hspace{1cm}} \text{ cm}^3$ | 28. $45\,000 \text{ mm}^3 = \underline{\hspace{1cm}} \text{ cm}^3$ | 29. $1.2 \text{ m}^2 = \underline{\hspace{1cm}} \text{ cm}^2$ |
| 30. $180 \text{ mL} = \underline{\hspace{1cm}} \text{ L}$ | 31. $7 \text{ m}^3 = \underline{\hspace{1cm}} \text{ cm}^3$ | 32. $9500 \text{ cm}^2 = \underline{\hspace{1cm}} \text{ m}^2$ |

CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

- | | |
|--|---|
| 33. A tank holds 2.4 m^3 . How many litres is that?
..... | 34. A jug holds 750 mL . How many cm^3 is that?
..... |
| 35. A room is 4 m by 3 m . Give its area in cm^2 .
..... | 36. A cube of side 10 cm holds how many mL ?
..... |
| 37. Explain why 1 m^2 is $10\,000 \text{ cm}^2$ and not 100 cm^2 .
..... | 38. A pool holds $30\,000 \text{ L}$. Give this in m^3 .
..... |
| 39. A sheet of paper is 21 cm by 30 cm . Give its area in mm^2 .
..... | 40. A medicine spoon holds 5 mL . How many spoons in 0.2 L ?
..... |

Worksheet 5: Volume and capacity

Week 3 · Worksheet 5 of 6 · 40 questions

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) · MA4-ARE-C-01 (area) · MA4-VOL-C-01 (volume and capacity) · MAO-WM-01

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Volume of a prism = area of the cross-section $\times$ length. 1 cm^3 holds exactly 1 mL.

EASY

20 QUESTIONS

Find the volume of each rectangular prism.

1. 10 cm by 4 cm by 6 cm _____
2. 5 m by 3 m by 2 m _____
3. 8 mm by 8 mm by 8 mm _____
4. 12 cm by 5 cm by 2 cm _____
5. 7 m by 4 m by 3 m _____
6. 6 cm by 6 cm by 6 cm _____
7. 20 mm by 10 mm by 5 mm _____
8. 9 m by 2 m by 4 m _____
9. 15 cm by 3 cm by 2 cm _____
10. 11 m by 5 m by 2 m _____

Find the volume of each triangular prism (base, height, length).

11. 8 m, 5 m, 12 m _____
12. 6 cm, 4 cm, 10 cm _____
13. 10 mm, 3 mm, 7 mm _____
14. 12 m, 6 m, 5 m _____
15. 9 cm, 8 cm, 4 cm _____
16. 14 mm, 5 mm, 6 mm _____
17. 4 m, 4 m, 9 m _____
18. 16 cm, 10 cm, 3 cm _____
19. 7 m, 2 m, 8 m _____
20. 20 mm, 6 mm, 5 mm _____

MEDIUM

12 QUESTIONS

Answer each question.

21. A cube has side 5 cm. Find its volume. _____
22. A cube has volume 64 cm^3 . Find its side. _____
23. A prism has cross-section 24 cm^2 and length 9 cm. Find its volume. _____
24. A tank is 2 m by 1.5 m by 1 m. Find its volume. _____
25. That tank's capacity in litres. _____
26. A box is 30 cm by 20 cm by 10 cm. Find its capacity in litres. _____
27. A prism has volume 180 m^3 and cross-section 20 m^2 . Find its length. _____
28. A cube has side 0.5 m. Find its volume in m^3 . _____
29. A fish tank is 60 cm by 30 cm by 40 cm. Find its capacity in litres. _____
30. A prism has volume 96 cm^3 and length 8 cm. Find its cross-section area. _____
31. A rectangular prism is 4 cm by 4 cm by 9 cm. Find its volume. _____
32. A container holds 5.5 L. Give its volume in cm^3 . _____

Worksheet 5: Volume and capacity

Week 3 · Worksheet 5 of 6 · 40 questions · continued

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) · MA4-ARE-C-01 (area) · MA4-VOL-C-01 (volume and capacity) · MAO-WM-01

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CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

1. A rectangular water tank is 2 m by 1.5 m by 1 m. How many litres does it hold?

.....

2. A swimming pool is 25 m by 10 m and 1.4 m deep. Find its capacity in litres.

.....

3. A cube has surface area 96 cm^2 . Find its volume.

.....

4. A prism has a triangular cross-section with base 8 m and height 5 m, and is 12 m long. Find its volume.

.....

5. A box 40 cm by 25 cm by 20 cm is filled with 1 cm cubes. How many fit?

.....

6. A tank holds 480 L. Its base is 1.2 m by 0.8 m. Find its depth.

.....

7. Explain why volume is measured in cubic units but area is not.

.....

8. A composite solid is a 10 by 4 by 3 cm block with a 4 by 4 by 3 cm block on top. Find its volume.

.....

Worksheet 6: Mixed revision

Week 3 · Worksheet 6 of 6 · 40 questions

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) · MA4-ARE-C-01 (area) · MA4-VOL-C-01 (volume and capacity) · MAO-WM-01

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This sheet mixes every skill from the Area and Volume chapter. Work through it as though it were the topic test.

EASY

20 QUESTIONS

Find the perimeter.

1. Rectangle 15 cm by 9 cm _____
2. Square, side 7 m _____
3. Triangle 5, 7, 8 cm _____
4. Rectangle 12 m by 4 m _____

Find the circumference (1 dp).

5. Radius 5 cm _____
6. Diameter 18 mm _____
7. Radius 3 m _____
8. Diameter 10 cm _____

Find the area.

9. Rectangle 8 cm by 5 cm _____
10. Triangle base 12 mm, height 9 mm _____
11. Parallelogram base 20 m, height 7 m _____
12. Circle radius 6 cm (1 dp) _____
13. Square side 9 m _____
14. Triangle base 10 cm, height 6 cm _____

Find the volume.

15. Prism 10 by 4 by 6 cm _____
16. Prism 5 by 3 by 2 m _____
17. Triangular prism: base 8 m, height 5 m, length 12 m _____
18. Cube of side 4 cm _____

Complete each conversion.

19. $3 \text{ m}^2 = \underline{\hspace{1cm}} \text{ cm}^2$
20. $200 \text{ mm}^2 = \underline{\hspace{1cm}} \text{ cm}^2$

MEDIUM

12 QUESTIONS

Answer each question.

21. $15 \text{ cm}^3 = \underline{\hspace{1cm}} \text{ mm}^3$ _____
22. $3 \text{ L} = \underline{\hspace{1cm}} \text{ mL}$ _____
23. $2500 \text{ mL} = \underline{\hspace{1cm}} \text{ L}$ _____
24. A rectangle has perimeter 34 cm, length 11 cm. Find the width. _____
25. A triangle has area 48 cm^2 , base 12 cm. Find the height. _____
26. Circle, diameter 14 cm: find the area (1 dp). _____
27. A tank 2 m by 1.5 m by 1 m: find its capacity in litres. _____
28. A cube has volume 125 cm^3 . Find its side. _____
29. A square has perimeter 36 m. Find its area. _____
30. A prism has cross-section 30 cm^2 and length 7 cm. Find its volume. _____
31. Circle, radius 4 m: find the circumference (1 dp). _____
32. A rectangle has area 96 m^2 and width 8 m. Find its length. _____



Worksheet 6: Mixed revision

Week 3 • Worksheet 6 of 6 • 40 questions • continued

NSW syllabus outcomes

MA4-LEN-C-01 (perimeter and circumference) • MA4-ARE-C-01 (area) • MA4-VOL-C-01 (volume and capacity) • MAO-WM-01

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● CHALLENGING

8 QUESTIONS

Answer each question. Show your working.

1. A rectangular lawn 12 m by 8 m has a 3 m square garden bed removed. Find the remaining area.

.....

2. Find the shaded area of an L-shape: a 12 cm by 9 cm rectangle with a 5 cm by 4 cm corner removed.

.....

3. A rectangular water tank is 2 m by 1.5 m by 1 m. How many litres does it hold?

.....

4. A bicycle wheel has diameter 66 cm. How far does it travel in 20 turns?

.....

5. A circular pond of radius 4 m is edged at \$22 per metre. Find the cost.

.....

6. A composite solid: a 10 by 4 by 3 cm block with a 4 by 4 by 3 cm block on top. Find its volume.

.....

7. Carpet costs \$45/m². A room is 4.5 m by 3.2 m. Find the cost.

.....

8. A prism has volume 180 m³ and cross-section 20 m². Find its length.

.....